Customer Shopping Behavior Analysis

1. Project Overview

This project focuses on understanding how customers interact with a retail platform by analyzing their purchase behavior across different categories, demographics, and engagement factors.

Instead of just looking at surface-level metrics, the goal was to break down *why* customers behave the way they do — what drives higher spending, which segments are more valuable, and how factors like discounts, subscriptions, and product categories influence decisions.

To approach this in a structured way, I followed a workflow similar to what is typically used in industry:

- Cleaning and preparing raw data using Python
- Structuring and querying data using SQL
- Building an interactive dashboard in Power BI to communicate insights

Business Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases. You are tasked with analyzing the company's consumer behavior dataset to answer the following overarching business question: "How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Deliverables

1. Data Preparation & Modeling (Python): Clean and transform the raw dataset for analysis

2. **Data Analysis (SQL):** Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.

3. Visualization & Insights (Power BI): Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.

4. **Report and Presentation**: Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.

5. **GitHub Repository**: Include all Python scripts, SQL queries, and dashboard files in a well- structured repository.

Dataset Summary

The dataset contains around 3900 transaction records capturing different aspects of customer behavior. Each row represents a purchase, along with associated customer and product details.

Key features included:

- Customer attributes like age, gender, and location
- Product-related details such as category and item purchased
- Transaction-level information including purchase amount and discounts
- Behavioral indicators like previous purchases and subscription status
- Customer feedback through review ratings

While exploring the dataset, I noticed that most of the fields were structured well, but there were missing values in the review rating column, which needed to be handled before analysis.

Sample Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Customer	Age	Gender	Item Purch	Category	Purchase A	Location	Size	Color	Season	Review Ra	Subscriptic	Shipping Ty	Discount A	Promo Co	Previous P	Payment M	Frequency of Pur	
2	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly	
3	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly	
4	3	50	Male	Jeans	Clothing	73	Massachus	S	Maroon	Spring	3.1	Yes	Free Shipp	Yes	Yes	23	Credit Car	Weekly	
5	4	21	Male	Sandals	Footwear	90	Rhode Isla	M	Maroon	Spring	3.5	Yes	Next Day A	Yes	Yes	49	PayPal	Weekly	
6	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipp	Yes	Yes	31	PayPal	Annually	
7	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	Venmo	Weekly	
8	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipp	Yes	Yes	49	Cash	Quarterly	
9	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipp	Yes	Yes	19	Credit Car	Weekly	
10	9	26	Male	Coat	Outerwear	97	West Virgi	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	Venmo	Annually	
11	10	57	Male	Handbag	Accessorie	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Ship	Yes	Yes	4	Cash	Quarterly	
12	11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pick	Yes	Yes	26	Bank Trans	Bi-Weekly	
13	12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pick	Yes	Yes	10	Bank Trans	Fortnightly	
14	13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes	Yes	37	Venmo	Fortnightly	
15	14	65	Male	Dress	Clothing	51	New Hamp	M	Violet	Spring	4.7	Yes	Express	Yes	Yes	31	PayPal	Weekly	
16	15	64	Male	Coat	Outerwear	53	New York	L	Teal	Winter	4.7	Yes	Free Shipp	Yes	Yes	34	Debit Card	Weekly	
17	16	64	Male	Skirt	Clothing	81	Rhode Isla	M	Teal	Winter	2.8	Yes	Store Pick	Yes	Yes	8	PayPal	Monthly	
18	17	25	Male	Sunglasses	Accessorie	36	Alabama	S	Gray	Spring	4.1	Yes	Next Day A	Yes	Yes	44	Debit Card	Bi-Weekly	
19	18	53	Male	Dress	Clothing	38	Mississippi	XL	Lavender	Winter	4.7	Yes	2-Day Ship	Yes	Yes	36	Venmo	Quarterly	
20	19	52	Male	Sweater	Clothing	48	Montana	S	Black	Summer	4.6	Yes	Free Shipp	Yes	Yes	17	Cash	Weekly	
21	20	66	Male	Pants	Clothing	90	Rhode Isla	M	Green	Summer	3.3	Yes	Standard	Yes	Yes	46	Debit Card	Bi-Weekly	
22	21	21	Male	Pants	Clothing	51	Louisiana	M	Black	Winter	2.8	Yes	Express	Yes	Yes	50	Cash	Every 3 Months	
23	22	31	Male	Pants	Clothing	62	North Car	M	Charcoal	Winter	4.1	Yes	Store Pick	Yes	Yes	22	Debit Card	Quarterly	
24	23	56	Male	Pants	Clothing	37	California	M	Peach	Summer	3.2	Yes	Store Pick	Yes	Yes	32	Debit Card	Annually	
25	24	31	Male	Pants	Clothing	88	Oklahoma	XL	White	Winter	4.4	Yes	Express	Yes	Yes	40	Credit Car	Weekly	
26	25	18	Male	Jacket	Outerwear	22	Florida	M	Green	Fall	2.9	Yes	Store Pick	Yes	Yes	16	Debit Card	Weekly	

Python Processing

Data Preparation and Processing (Python)

Before jumping into analysis, I focused on making the dataset reliable and easier to work with.

I started by loading the data using pandas and checking its structure using `.info()` and `.describe()` to understand data types, ranges, and inconsistencies.

One issue I encountered was missing values in the review rating column. Instead of dropping those rows, I handled them carefully so that I didn't lose useful data.

To make the analysis more meaningful, I created additional features:

- Converted raw age values into grouped segments like young adult, middle-aged, etc., so comparisons become clearer
- Used previous purchase data to later segment customers into different behavioral groups

I also standardized column names and checked for consistency across categorical variables to avoid issues during SQL querying.

```
[13]: # Imputing missing values in Review Rating column with the median rating of the product category
      df['Review Rating']=df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))

[14]:
      df.isnull().sum()

[14]: Customer ID               0
      Age                       0
      Gender                    0
      Item Purchased            0
      Category                  0
      Purchase Amount (USD)     0
      Location                  0
      Size                      0
      Color                     0
      Season                    0
      Review Rating             0
      Subscription Status       0
      Shipping Type             0
      Discount Applied          0
      Promo Code Used           0
      Previous Purchases        0
      Payment Method            0
      Frequency of Purchases    0
      dtype: int64
```

```
[32]: # create a new column age_group

      labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
      df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)

[36]: df[['age','age_group']].head(10)
```

[36]:

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

Cleaned Dataset

```
[53]: df.head(30)
```

[53]:

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_type
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Ai
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping
5	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standarc
6	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping
7	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping
8	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express
9	10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping
10	11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pickup

SQL Analysis

Data Analysis using SQL

Once the data was cleaned, I moved it into a structured SQL environment to analyze it from a business perspective.

Instead of random queries, I framed specific business questions and answered them systematically.

Revenue by Gender

I started by comparing how much revenue is generated by male vs female customers. This helps in understanding if one segment contributes more financially.

High-Spending Discount Users

I wanted to check whether discounts are only attracting low-value purchases or if high-value customers are also using them.

Top Rated Products

Customer reviews often reflect satisfaction, so I identified products with the highest average ratings.

I continued the analysis by digging deeper into behavioral and operational patterns.

- Compared average spending between shipping types to see if faster delivery correlates with higher purchase value
- Analyzed subscription vs non-subscription behavior to understand long-term engagement

- Identified products heavily influenced by discounts, which can indicate price sensitivity
- Segmented customers into new, returning, and loyal groups based on their purchase history
- Identified top-performing products within each category
- Checked whether repeat buyers are more likely to subscribe
- Calculated revenue contribution across different age groups

This step helped in moving from raw data to structured insights that can directly support decision-making.

```sql
--Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?
SELECT subscription_status,
       COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;

--Q10. What is the revenue contribution of each age group?
SELECT
    age_group,
    SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue desc;
```

Query Query History

```sql
--Q1. What is the total revenue generated by male vs. female customers?
select gender, SUM(purchase_amount) as revenue
from customer
group by gender


--Q2. Which customers used a discount but still spent more than the average purchase amount?
select customer_id, purchase_amount
from customer
where discount_applied = 'Yes' and purchase_amount >= (select AVG(purchase_amount) from customer)


-- Q3. Which are the top 5 products with the highest average review rating?
select item_purchased, round(avg(review_rating::numeric),2) as "Average Product Rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5

--Q4. Compare the average Purchase Amounts between Standard and Express Shipping.
select shipping_type,
ROUND(AVG(purchase_amount),2)
from customer
where shipping_type in ('Standard','Express')
group by shipping_type;
```

Snipping

Query Query History

```sql

--Q5. Do subscribed customers spend more? Compare average spend and total revenue
--between subscribers and non-subscribers.
SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount),2) AS avg_spend,
       ROUND(SUM(purchase_amount),2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue,avg_spend DESC;


--Q6. Which 5 products have the highest percentage of purchases with discounts applied?
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)/COUNT(*),2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

Query Query History

```sql
--Q7. Segment customers into New, Returning, and Loyal based on their total
-- number of previous purchases, and show the count of each segment.
with customer_type as (
SELECT customer_id, previous_purchases,
CASE
    WHEN previous_purchases = 1 THEN 'New'
    WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
    ELSE 'Loyal'
    END AS customer_segment
FROM customer)

select customer_segment,count(*) AS "Number of Customers"
from customer_type
group by customer_segment;

--Q8. What are the top 3 most purchased products within each category?
WITH item_counts AS (
    SELECT category,
           item_purchased,
           COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank
    FROM customer
    GROUP BY category, item_purchased
)
SELECT item_rank,category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <=3;
```

Key Insights

After analyzing the data, a few patterns stood out clearly.

One of the strongest observations was that **young adults contribute the highest revenue**, which makes them the most valuable customer segment. This suggests that marketing and product strategies should be aligned more closely with this group's preferences.

Another interesting finding was that **non-subscribers generate higher total revenue compared to subscribers**. This indicates that the current subscription model may not be delivering enough value to attract or retain high-spending users.

Looking at product performance, the **clothing category consistently drives the highest sales**, which suggests strong demand and possibly better product-market fit in that segment.

Discount analysis showed that certain products are **highly dependent on discounts**, which can be a double-edged sword — good for volume, but risky for margins.

Customer segmentation revealed that a large portion of users fall into the **loyal category**, which is a positive signal for retention, but also an opportunity to further increase lifetime value.

Dashboard

Dashboard Visualization (Power BI)

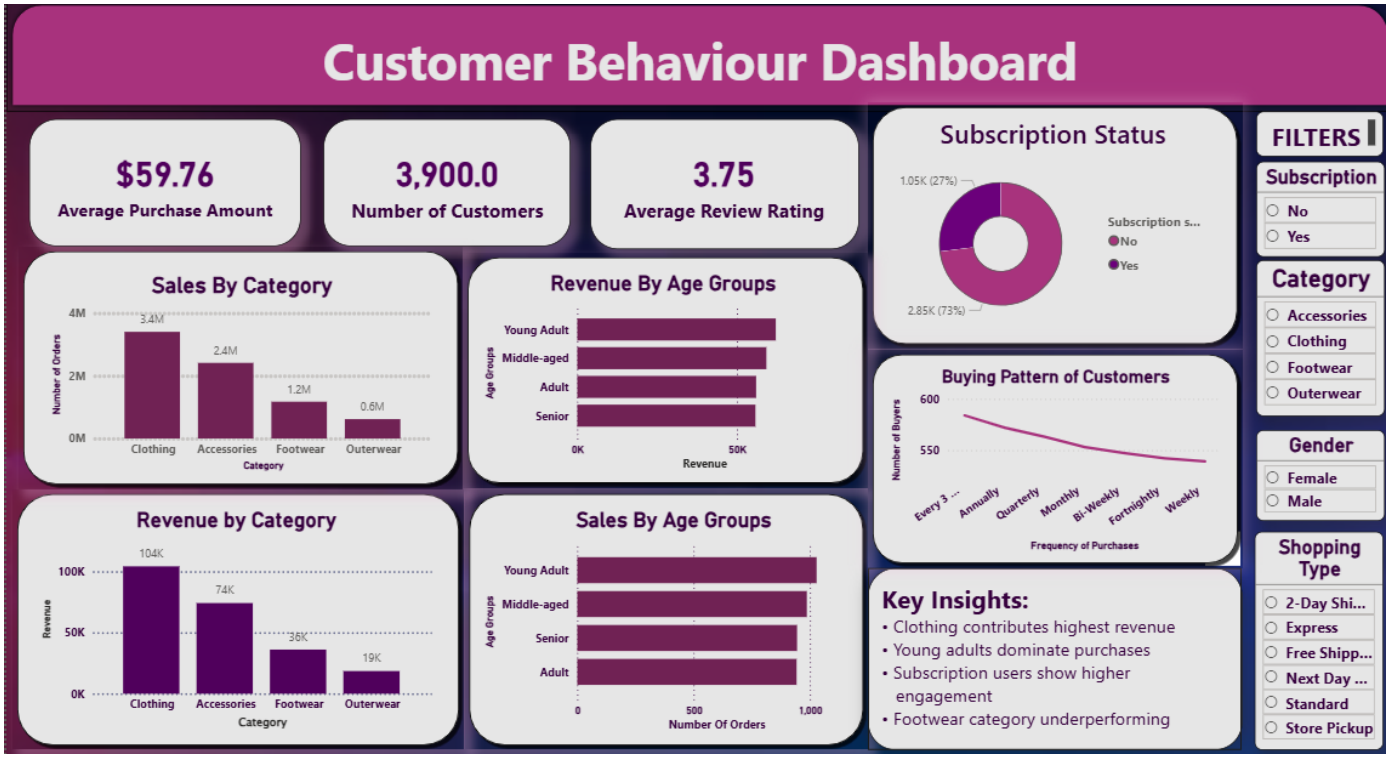
To bring all these insights together, I built an interactive dashboard in Power BI.

The idea was not just to present numbers, but to make the data easy to explore and interpret.

The dashboard includes:

- KPI cards for quick overview of key metrics
- Category-wise and age-group-wise breakdowns
- Subscription analysis through visual comparisons
- Product-level insights to identify top performers

This allows users to quickly move from high-level metrics to detailed breakdowns without switching tools



Business Recommendations and Future Scope

Based on the analysis, a few clear actions can be taken.

First, there is a strong case for **reworking the subscription model**. Since non-subscribers are contributing more revenue, the company needs to rethink what value it is offering through subscriptions. This could include exclusive discounts, faster shipping, or personalized recommendations.

Second, the company should **focus more on high-performing segments like young adults**. This can be done through targeted campaigns, curated product offerings, or personalized experiences.

Third, while discounts are effective, they should be used more strategically. Instead of applying them broadly, the company can **target price-sensitive products and segments**, ensuring that margins are not unnecessarily reduced.

From a product perspective, categories like clothing can be further expanded, while underperforming categories may need repositioning or better marketing.

Looking ahead, this analysis can be extended by incorporating time-based data to track trends over months or seasons, which would make the insights even more actionable.

Additionally, integrating customer-level lifetime value and behavioral tracking can help move from descriptive analysis to predictive decision-making.